

ResQGrid

Final Complete Phase-Wise Plan (v4)

Every Feature Mapped · Sub-Divided Phases · Research Gaps · ESP32 Offline-Node Device

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1. How to Read This Plan

Har phase ab chhote 'Parts' (Part A, Part B...) mein divide kiya gaya hai, taaki samajhna aasan ho. Tags: orange = Research Gap, green = Key Feature, blue = Core Feature (Not Optional).

2. Complete Feature Checklist (Overview)

Ye sab kuch is document mein cover hoga — poori conversation mein discuss hui har cheez:

- Citizen: SOS-submission (Rescue/Food/Water/Medicine/Medical), photo, GPS, status-tracking
- Nearby-device detection (BLE scan, jab internet na ho)
- BLE Store-and-Forward relay (Citizen → nearby phone → jab internet mile → Cloud)
- ResQGrid Offline-Node (ESP32 device) — Rescue-Team WiFi-hotspot jisse Citizen phone directly connect ho
- Device ka local server, SOS-queue, local-storage, captive-portal
- LoRa node-to-node extension (jab WiFi-range khatam ho jaaye)
- CADERF: Normalization, Relationship-detection (Supports/Conflicts/Updates/Duplicates/Completes)
- Duplicate-Filtering Bottleneck (Rescue-Team ko sirf clean incidents dikhna)
- Incident Timeline (chronological history)
- Local Knowledge Base (10 categories)
- ResQ App roles: Volunteer, Hospital, NGO, Camp, Rescue-Team
- Government Dashboard: GIS-map, Heatmap, Incident-list, Resource-panel
- Real-time updates (Socket.io)
- Simulation & 4-method Evaluation (Raw/Majority-Vote/Weighted-Fusion/CADERF)
- Testing, Documentation, Presentation

PHASE 1 — Setup (2-3 Days)**Part A — Project Structure**

- GitHub repo banao, folders: server/, resqgrid_citizen_app/, resqgrid_resq_app/

Part B — Backend Base

- Node.js init, Express install, MongoDB Atlas free-cluster
- .env file (PORT, MONGO_URI, JWT_SECRET)

Deliverable: Empty skeleton running, first commit.

PHASE 2 — Database Design (3-4 Days)**Part A — Core Collections**

- User (role: Citizen/Volunteer/Hospital/NGO/Camp/RescueTeam/Admin)
- EvidenceReport (source, location, timestamp, evidenceType, mediaURL, status)

Part B — Incident & Knowledge Collections

- Incident (estimatedVictims, confidenceScore, needs, status, timeline[])
- LocalKnowledge (category, location, description, submittedBy, lastUpdated)
- ResourceInventory (ownerType, resourceType, quantity)

Part C — Test Data

- Seeder script — dummy villages, camps, hospitals

★ **GAP 2 — Unified Evidence Data Model: EvidenceReport schema hi Citizen/Volunteer/Hospital/Drone sabka data ek common structure mein store karta hai.**

Deliverable: Saare models MongoDB mein working, test-data visible.

PHASE 3 — Backend Core APIs (5-6 Days)**Part A — Authentication**

- JWT-based login, role-based middleware (7 roles)

Part B — Core CRUD APIs

- /api/sos, /api/volunteer-update, /api/hospital-update, /api/ngo-inventory, /api/camp-status, /api/knowledge

Deliverable: Postman se saare APIs verified.

PHASE 4 — CADERF — Evidence Reconciliation Engine (7-8 Days)

Ye poore project ka core research-module hai.

Part A — Normalization

- Har source (Citizen/Volunteer/Hospital/Drone) ka data common-format mein convert

Part B — Relationship Detection

- Supports / Conflicts / Updates / Duplicates / Completes — in 5 relationships ko identify karne wala logic

Part C — Reconciliation + Duplicate-Filtering Bottleneck

- Duplicate-removal → conflict-flagging → context-weighted-merge → confidence-score

★ **GAP 1 — DISASTER-SPECIFIC EVIDENCE RECONCILIATION — Sabse important, highest-confidence research-contribution.**

🔑 **KEY FEATURE — Duplicate-Filtering Bottleneck — Rescue-Team tak pahunchne se pehle saare duplicate-reports filter ho jaate hain. Rescue-Team ko sirf ek clean 'Unified Incident' dikhta hai, 100 raw-messages nahi.**

Part D — Incident Generation

- Reconciled evidence se Unified-Incident record banta hai

Deliverable: 5 conflicting test-reports do, ek clean deduplicated Incident nikal ke dikhao.

PHASE 5 — Incident Timeline + Local Knowledge Base (3-4 Days)**Part A — Incident Timeline**

- Har Incident ke saath timestamped-history (08:01 Citizen → Need Rescue, 08:10 Volunteer → Reached, etc.)

Part B — Local Knowledge — 10 Categories

- Road, Bridge, Shelter, Boat-Launch-Point, Water-Depth, Village-Contact, Local-Language, Medical-Resource, Livestock-Area, Danger-Zone
- Entries kabhi delete nahi hoti, sirf update hoti hain

★ **GAP 3 — PERSISTENT LOCAL DISASTER KNOWLEDGE — Outside-responders ko agle-disaster mein turant available.**

Deliverable: Incident-timeline aur Village-knowledge database mein dikhna chahiye.

PHASE 6 — Citizen App (Flutter) (5-6 Days)**Part A — Core Screens**

- Login/Splash, Home (Need-Help grid), SOS-submission (category, photo, auto-GPS)

Part B — Status & Info Screens

- SOS-status tracking (Sent → Relayed → Delivered → Acknowledged)
- Nearby Camp/Hospital view, Local-Knowledge view

Deliverable: Citizen SOS bhej sake, status track kar sake.

PHASE 7 — ResQ Multi-Role App (Flutter) (5-6 Days)

Ek hi codebase, role-based-login ke baad alag dashboard khulta hai:

Part A — Volunteer Role

- Status-update, Local-Knowledge submission, Nearby-SOS list

Part B — Hospital Role

- Beds/Doctors/Blood/Medicine/Oxygen update

Part C — NGO Role

- Inventory-management (Food/Water/Blankets/Generators/Medicine)

Part D — Camp Role

- Capacity + Resource-status update

Part E — Rescue-Team Role

- Assigned-Incident list (clean, deduplicated), Local-Knowledge panel, Boat-tracking

Deliverable: Har role apna dashboard use kar sake, ek hi codebase se.

PHASE 8 — BLE Offline-Relay (Core Feature) (5-6 Days)**Part A — Nearby-Device Detection**

- flutter_blue_plus se background-scan — jab internet na ho, nearby phone (jisme app installed hai) automatically detect ho

Part B — Store & Forward Logic

- Citizen ka SOS local-queue mein save → nearby-device mile to copy → jab kisi bhi device ko internet mile → Cloud-sync

Part C — Communication Abstraction Layer

- Priority-order: Internet available? → Direct. Nahi? → BLE-nearby-device dhoondo → Relay. Kuch nahi? → Local-queue mein wait.

CORE (Not Optional) — BLE Offline-Relay ye 'ResQGrid' ke 'Offline-First' naam ka asal matlab hai — Citizen App aur ResQ App dono ke andar built-in hai.

Deliverable: 2-phone live-demo — internet-off karke SOS relay hote dikhao.

PHASE 9 — ResQGrid Offline-Node Device (ESP32 Hardware) (6-7 Days)

Ye ek portable hardware-device hai jo Rescue-Team apne saath carry karti hai — Citizen ke phone ko bina kisi extra-app ke iske WiFi se seedha connect hone deta hai.

Part A — Basic Prototype (Pehle Ye Karo)

- ESP32 board (₹300-700), SoftAP-mode configure karo (Wi-Fi name: 'ResQGrid-SOS')

- Local HTTP-server chalao ESP32 ke andar (192.168.4.1 par)
- SOS-receive API + Local-storage (agar Rescue-Team-phone temporarily disconnect ho, SOS save rahe)

Part B — Citizen-Side Experience

- Citizen phone WiFi-list mein 'ResQGrid-SOS' dekhta hai → connect karta hai (koi extra-hardware nahi chahiye)
- Captive-portal — automatically 'RESQGRID EMERGENCY' page khulta hai, IP manually type nahi karna padta

Part C — Advanced Additions (Agar Time Bache)

- External-antenna (range-improvement), Battery+BMS, Status-LEDs, SOS/Reset buttons
- Waterproof IP65-enclosure (sabse aakhir mein, sirf agar prototype poora working ho chuka ho)

 **KEY FEATURE — Citizen-Side Zero-Hardware —** Ye device WiFi (na ki LoRa) use karta hai, isliye normal Citizen-phone bina kisi dongle/app-change ke seedha connect ho sakta hai. Range realistically 30-100m hogi (external-antenna se thodi zyada), 200-300m jaisa commercial-claim nahi karenge bina field-testing ke.

Deliverable: 1 working ESP32-prototype — phone se WiFi-connect karke SOS bhejo, device receive karke store kare.

PHASE 10 — LoRa Extension (Optional, Node-to-Node) (3-4 Days)

LoRa ab Citizen ke liye mandatory nahi hai — ye sirf tab kaam aata hai jab do ResQGrid-Nodes (Phase 9 wale devices) ek-dusre se WiFi-range se bahar hon.

Part A — Hardware

- 1x extra ESP32+LoRa board (₹1,770-2,500), 865-867 MHz India-band

Part B — Node-to-Node Bridge

- Citizen-Phone → WiFi → ResQGrid-Node-A → LoRa → ResQGrid-Node-B → WiFi → Rescue-Team-Phone

Optional-Bonus — sirf Phase 1-9 complete hone ke baad, agar time/budget bache tabhi shuru karo.

Deliverable: 2 ResQGrid-Nodes ke beech ek test-message LoRa se transfer hote dikhao.

PHASE 11 — Government Dashboard (React Web) (5-6 Days)

Part A — Map & Visualization

- Leaflet Map — SOS/Camp/Hospital markers, Heatmap layer

Part B — Panels

- Incident-list panel (deduplicated), Resource-status panel, Analytics

Deliverable: Working web-dashboard, live data ke saath.

PHASE 12 — Real-Time Layer (Socket.io) (2-3 Days)

- Backend Socket.io setup, naya SOS/Incident bante hi dashboard auto-update (bina refresh)

Deliverable: SOS bhejte hi dashboard turant update ho.

PHASE 13 — Simulation & Evaluation (4-5 Days)

Part A — Data Generation

- 1000-report dummy dataset generator

Part B — Comparison

- 4 methods: Raw-Reports / Majority-Vote / Weighted-Fusion-Baseline / CADERF
- Metrics: duplicate-rate, response-time, accuracy

★ GAP 4 — EXPERIMENTAL EVALUATION OF DISASTER EVIDENCE FRAMEWORKS.

Deliverable: Result-table/chart, directly research-paper mein use hoga.

PHASE 14 — Testing & Polish (4-5 Days)

- End-to-end testing, UI-polish, error-handling, edge-cases

PHASE 15 — Documentation & Presentation (3-4 Days)

- README, PPT, Demo-script (BLE + ESP32-Node scenes), Research-paper draft

18. Master Feature-Coverage Table

Function/Feature	Phase	Tag
Citizen SOS submission (photo, GPS, category)	Phase 6	—
Nearby-device detection (BLE scan)	Phase 8	● Core
Store & Forward relay (phone-to-phone)	Phase 8	● Core
ResQGrid Offline-Node (ESP32 WiFi-AP device)	Phase 9	Hardware
Device local-server, SOS-queue, captive-portal	Phase 9	Hardware
LoRa node-to-node bridge	Phase 10	Optional
Evidence Normalization	Phase 4	★ Gap 2 (schema base)
Relationship Detection (5 types)	Phase 4	★ Gap 1
Duplicate-Filtering Bottleneck	Phase 4	🔑 Key Feature
Incident Timeline	Phase 5	—
Local Knowledge Base (10 categories)	Phase 5	★ Gap 3
Volunteer / Hospital / NGO / Camp / Rescue-Team apps	Phase 7	—
Government Dashboard (GIS, Heatmap)	Phase 11	—
Real-time updates	Phase 12	—
Simulation + 4-way comparison	Phase 13	★ Gap 4

19. Master Research-Gap Table

#	Research Gap	Phase
1	Disaster-Specific Evidence Reconciliation	Phase 4
2	Unified Disaster Evidence Data Model	Phase 2
3	Persistent Local Disaster Knowledge Representation	Phase 5
4	Experimental Evaluation of Disaster Evidence Frameworks	Phase 13

'Policy → Software Translation' Motivation hai, alag Research-Gap/Phase nahi — ye poore project ka overall justification hai (Introduction section).

20. Timeline, Team-Split, Cost

Phase Group	Approx Days
Setup + Database (Phase 1-2)	~6 din
Backend + CADERF + Knowledge (Phase 3-5)	~18 din
Apps (Citizen + ResQ) + BLE (Phase 6-8)	~15-17 din
ESP32-Node + LoRa (Phase 9-10)	~9-11 din
Dashboard + Real-time (Phase 11-12)	~7-9 din

Evaluation + Testing + Docs (Phase 13-15)		~11-14 din
Total (4-member team)		~70-75 din (~2.3-2.5 mahine)
Member	Responsibility	
Member 1	Backend + CADERF (Phase 2-5)	
Member 2	Citizen App + ResQ App + BLE (Phase 6-8)	
Member 3	Dashboard + Real-time (Phase 11-12)	
Member 4	ESP32-Node + LoRa + Simulation (Phase 9-10, 13)	
Item		Cost
Core software (Flutter, Node, MongoDB, GitHub)		₹0
BLE (built into Flutter apps)		₹0
ESP32 board (basic prototype)		₹300-700
LoRa extension board (optional)		₹1,770-2,500
Total Realistic Budget		₹300-700 (core+device) + ₹1,770-2,500 (if LoRa added)